

Priya's work when solving the equation $2x^2 - x - 20 = 2x$ using the quadratic formula is shown. Use Priya's work to answer questions 1 and 2.

$2x^2 - x - 20 = 2x$ $2x^2 - 3x - 20 = 0, \text{ so } a = 2, b = -3, c = -20$ $x = \frac{-3 \pm \sqrt{(-3)^2 - 4(2)(-20)}}{2(2)}$ $x = \frac{-3 \pm \sqrt{9 + 160}}{4}$ $x = \frac{-3 \pm \sqrt{169}}{4}$ $x = \frac{-3 \pm 13}{4}$ $x = \frac{10}{4} \text{ or } x = \frac{-16}{4}$ $x = \frac{5}{2} \text{ or } x = -4$	$-b = -(-3)$ $x = \frac{3 \pm \sqrt{9 + 160}}{4}$ $x = \frac{3 \pm \sqrt{169}}{4}$ $x = \frac{3 \pm 13}{4}$ $x = \frac{3 + 13}{4} = \frac{16}{4} = 4$ $x = \frac{3 - 13}{4} = \frac{-10}{4} = -\frac{5}{2}$ $x = -\frac{5}{2} \text{ or } x = 4$
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1. Circle the step where Priya made an error.
2. Correct Priya's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Solve each equation by using the quadratic formula. Show your work.

3. $2x^2 + 3x - 20 = 0$ $a = 2, b = 3, c = -20$

$$x = \frac{-3 \pm \sqrt{9 + 160}}{4}$$

$$x = \frac{-3 \pm 13}{4}$$

$$x = \frac{-3 + 13}{4} = \frac{10}{4} = \frac{5}{2}$$

$$x = \frac{-3 - 13}{4} = \frac{-16}{4} = -4$$

$$x = \frac{5}{2} \text{ or } x = -4$$

$$4. \quad 5x^2 + 4 = -9x \quad a = 5, b = 9, c = 4$$

$$5x^2 + 9x + 4 = 0$$

$$x = \frac{-9 \pm \sqrt{81 - 80}}{10}$$

$$x = \frac{-9 \pm \sqrt{1}}{10}$$

$$x = \frac{-9 + 1}{10} = \frac{-8}{10} = -\frac{4}{5}$$

$$x = \frac{-9 - 1}{10} = \frac{-10}{10} = -1$$

$$x = -\frac{4}{5} \text{ or } -1$$

$$5. \quad 0 = 2x^2 - x - 36 \quad a = 2, b = -1, c = -36$$

$$x = \frac{1 \pm \sqrt{1 + 288}}{4}$$

$$x = \frac{1 \pm \sqrt{289}}{4}$$

$$x = \frac{1 + 17}{4} = \frac{18}{4} = \frac{9}{2}$$

$$x = \frac{1 - 17}{4} = \frac{-16}{4} = -4$$

$$x = \frac{9}{2} \text{ or } -4$$

$$6. \quad 2x^2 - 2x - 3 = 5x \quad a = 2, b = -7, c = -3$$

$$2x^2 - 7x - 3 = 0$$

$$x = \frac{7 \pm \sqrt{49 + 24}}{4}$$

$$x = \frac{7 \pm \sqrt{73}}{4}$$

$$7. \quad x^2 - 8x + 10 = -6 \quad a = 1, b = -8, c = 16$$

$$x^2 - 8x + 16 = 0$$

$$x = \frac{8 \pm \sqrt{64 - 64}}{2}$$

$$x = \frac{8 \pm 0}{2}$$

$$x = 4$$

$$8. \quad -7 = 2x^2 + 10x \quad a = 2, b = 10, c = 7$$

$$2x^2 + 10x + 7 = 0$$

$$x = \frac{-10 \pm \sqrt{100 - 56}}{4}$$

$$x = \frac{-10 \pm \sqrt{44}}{4}$$

$$x = \frac{-10 \pm 2\sqrt{11}}{4}$$

$$x = \frac{-5 \pm \sqrt{11}}{2}$$

$$9. \quad 6x^2 + 24x - 30 = 0 \quad a = 6, b = 24, c = -30$$

$$x = \frac{-24 \pm \sqrt{576 + 720}}{12}$$

$$x = \frac{-24 \pm \sqrt{1,296}}{12}$$

$$x = \frac{-24 \pm 36}{12}$$

$$x = \frac{-24 + 36}{12} = \frac{12}{12} = 1$$

$$x = \frac{-24 - 36}{12} = \frac{-60}{12} = -5$$

$$x = 1 \text{ or } -5$$

$$10. \quad x + 3 = x^2 \quad a = 1, b = -1, c = -3$$

$$x^2 - x - 3 = 0$$

$$x = \frac{1 \pm \sqrt{1 + 12}}{2}$$

$$x = \frac{1 \pm \sqrt{13}}{2}$$